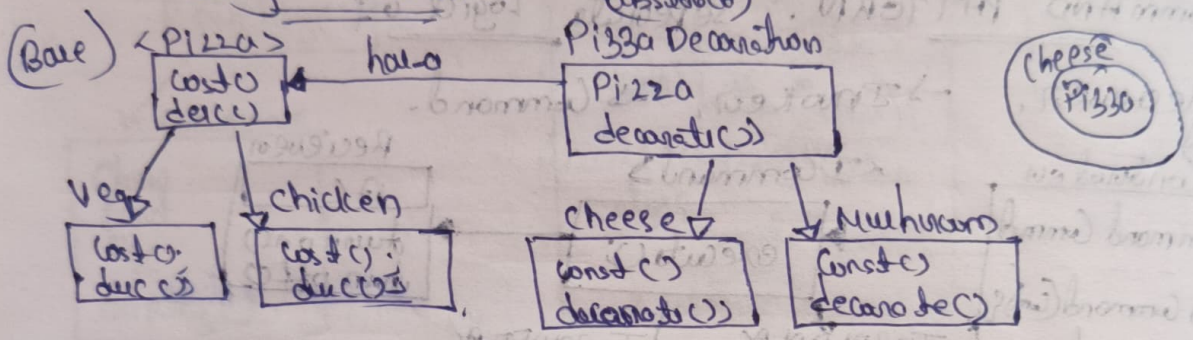
(18) Apply Coupons on Shopping Cart. DECORATOR PATTERN

→ Apply Coupons on the all Products in Cart which are eligible & get the final amount

→ Decorator Pattern :- Structural pattern allows to add the new functionality to object dynamically without modifying structure.

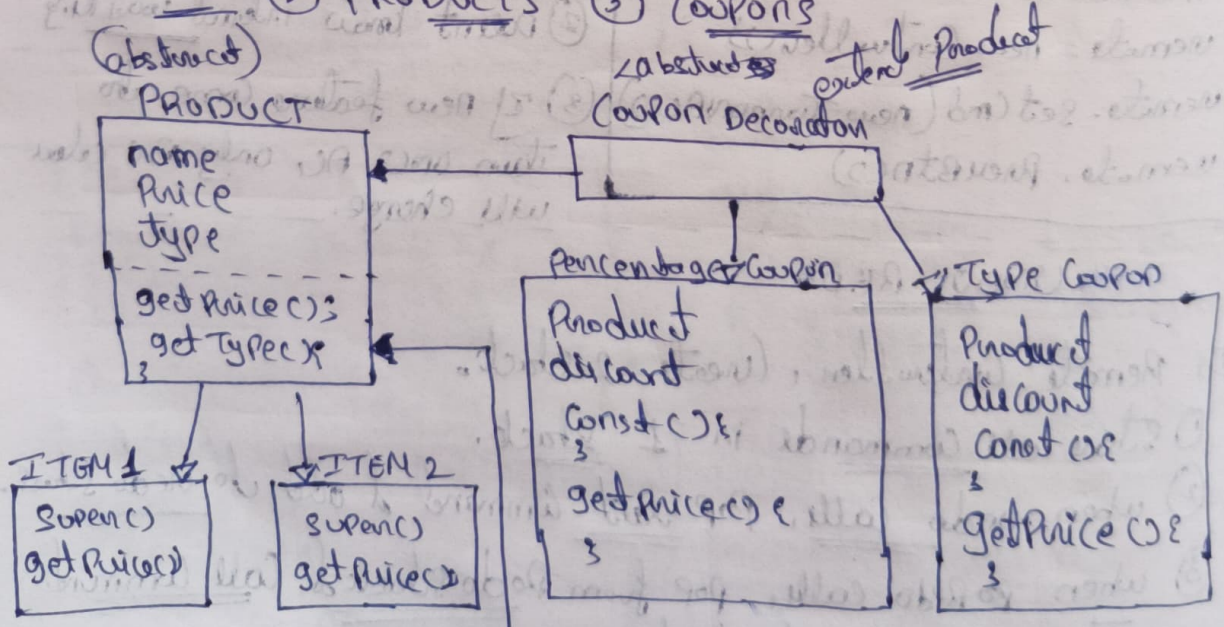



→ Note :- After all decorate we at end we get Pizza only with extra function.

---

COUPONS ON CART.

① CART ② PRODUCTS ③ COUPONS




SHOPPING CART

```
List<Product>
addToCart(Product){
  Product = new TypeCoupon(new
      Percent(Product, 10), 3, Type);
```

```
Main(){
item1, item2
shopingCart
add(1)
getPrice()
```

① Initially, when we create Product it has some price

② when we add it to cart, it pulls all the coupons decorators & finally it gets updated with discount price

③ Then we can get total price.

---

addToCart(Product)?

Product discountProd = new Type_Coupon(

new Percentage_Coupon(Product, 10.0), type());

list.add(discountProd);

Execution order → Inner will execute first

① getType() is called

② new Percentage_Coupon(Prod, 10.0) called.

③ new Type_Coupon(...) called.

→ Inside-out execution will happen.

---